

Emergent Vehicle Tracking System Using IR Sensor

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Abstract— To propose a signal track division that use universal position method to find the exact position of an emergency van (ex. Ambulance, Fire exchange) and remote users get the information transmitted using global position system which also provides traffic signal track monitoring system i.e., during object motion IR sensors detect the ambulance, pass the message to main controller which reflects the LED bulb to glow leading to closing of all the signals track, and immediately the emergent vehicle pass the signal track. The main principle of this scheme is to plan an emergent tracking method. The proposed system is implemented based on real time embedded system for this implementation we used ATMEG328P controller which is the heart of the system. The IR sensor is used to sense the data which is processed by the controller and sent the data to emergent tracking system through GPS.

Keywords— *Controller (ATMEG 328P), Zigbee, GSM, HTTPS protocol, Smart Traffic signal, Security, IR Sensor.*

I. INTRODUCTION

Projected plan is commercial, undependable and have the task of perfect ambulance track. Once the number of vehicle was increase in excess of the signal track, the Traffic police often found it hard to keep path of what was incident in the pointer. They involve a performance to make a decision somewhere each purpose be at within the least set instance and for the way long it opens the signal and conjointly machine van track area unit desired. GSM and emergency based track structure will provide helpful, valid time motor vehicle position and coverage to the controller. A GSM base track method will report to anywhere your motor vehicle is and where and how extended it has be. [1]. the method uses geographic location and instant in sequence from the Global Positioning Satellites. This scheme consists of assemble unit which implement on the motor vehicle to be rectify. It includes Global system for mobile communication, GPS receiver, as well as controller It can give traffic indication-monitor and managing method for inter-cities moving vehicle such as Ambulances, Fire extinguish [2]. For the period of motor vehicle movement, its real-time parameter such as gesture position is report by communication. This methodology has improvement of wireless machinery that gives dominant transfer management

pointer. [3]. the create use of GSM and GPS technology permit the arrangement to follow car and supply the foremost up-to-date in sequence regarding constant vehicle. This theme finds connectedness in real purpose interchange observation. It might be use as a expensive device for valid moment in time motor vehicle in sequence, overcrowding monitor, as well as method evaluation [4].

A smart, mechanized motor vehicle recognition method can determine the issues such as uncertain go past of signal routine, inappropriate operation of indicator point as well as capital, behind schedule arrival as per agenda, and people frustration. This container guide to improved passage run model and a improved accepting of interchange pointer behavior. This paper Include a variety of features similar to original, effortlessness of devise and simple achievement. It is completely incorporated in order that only once it's implement altogether signal then it's easy to follow car anyplace anytime [7].

Hardware illustration for track technique is shown in Fig one. Technique contains necessary presentation ATMEG 328DIP, a GPS, and GSM modem and taken as a whole method be a feature of close to the interchange gesture [6]. A track method will give efficient valid time go beyond motor vehicle information and track method motivation modernize position of the motor vehicle as well as wherever it have be, at what distance it is actually located. Observe and pass on the location information. Control module forms the vital part of the track element, which acquire and appraise the position data from the GPS unit. The GPS phone of reaction fraction of the car receive further as resolve the direction-finding announcement transmit by GPS position satellites, transform it keen on the GSM communication produced by GSM message checker, and send the communication to monitor heart via the GSM system [5].

II. LITERATURE SURVEY

[1] Which incorporates the trendy estimates of the urban and rural populations of 233 countries? Population

censuses are the most commonly used sources of data, although estimates obtained from population registers or administrative statistics. Classifying an area as urban may be based: a minimum population threshold; population density; proportion employed in non-agricultural sectors; the presence of infrastructure such as paved roads, electricity, piped water or sewers; and the presence of education or health services. In compiling information on city population size, the Population Division has endeavoured to use data or estimates based on the concept of urban agglomeration.

[2] Propose a reservation based system for alleviating traffic congestion, specifically at intersections, and under the assumption that the cars are controlled by agents. a specific metric for comparing the excellent of traffic manipulate at an intersection. This reservation-based system can perform two to three hundred times better than traffic lights and it can smoothly handle much heavier traffic conditions. This machine very carefully approximates an overpass, that's the most beneficial answer for the hassle.

Current methods for controlling traffic, specifically at intersections, will not be able to take advantage of the increased sensitivity and precision of autonomous vehicles as compared to human drivers. [3] Describe an autonomous intersection management system. Drivers and intersections on this mechanism are handled as self sufficient dealers in a multi agent system. In this multi agent system, intersections use a new reservation-based approach built around a detailed communication protocol.

[4] Present a framework designed initially for cyber cars (fully automated cars) but that could also be applied - though with major differences -to human driven cars. It is a world where vehicles have to reserve pieces of roads to cross a junction. This work is an enhancement of a previous work that demonstrated the feasibility of such a reservation algorithm.

[5] Designs and evaluates a reservation-based approach to intersection control that is designed to take full advantage of the unprecedented connectivity that the connected vehicle initiative promises to provide. To design and evaluate the "intelligent intersection," a novel simulation test bed for connected vehicle applications is developed. [8] Propose a novel "smart parking" system for an urban environment. The system assigns and reserves an optimal parking space based on the driver's cost function that combines proximity to destination and parking cost. Solves a mixed-integer linear programming.

III. PROPOSED ARCHITECTURE

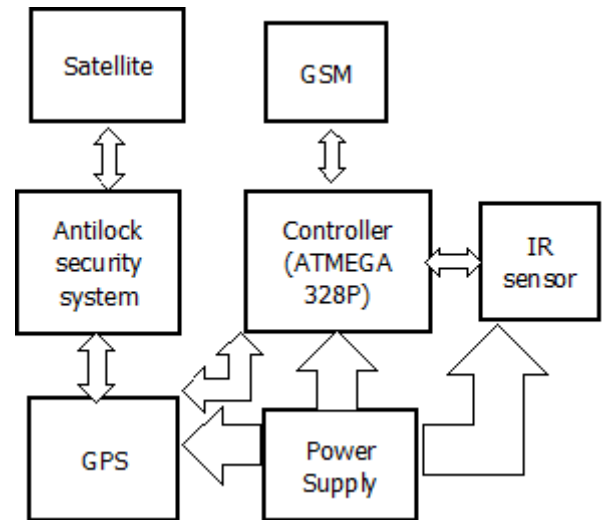


Fig.1 Block diagram of track system GSM HARDWARE

The central a part of specifics announcement regarding this methodology lays in wireless message visual show unit with the principle of use GSM unit to specific moment-uncontrollable-breathing house in sequence. It supports information of AT guidelines [10]. SIM900 be integral with an extensive variety of application.

SIM900 could be a Tri-gang GSM/GPRS part to work on the frequencies DCS 850 megacycle per second, EGSM 900 megacycle per second, as well as. PCS 1900 MHz. SIM900 facilitates GPRS multi-slot category ten capabilities and it conjointly supports the GPRS code then still as CS-4 with a small arrangement of 40mm x 33mm x 2.85 mm, SIM900 can grip approximately the entire whole condition in our function. Hence the CU can be associated with GSM module especially well throughout sequential interface [8]. GSM modem is as shown in Fig. 2.

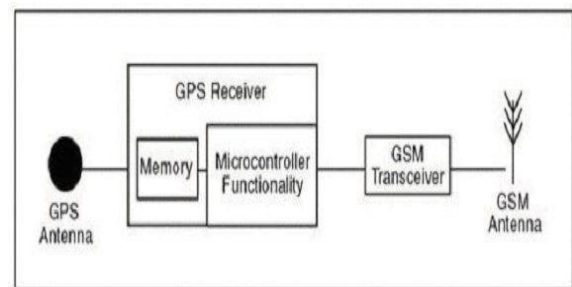


Fig. 2 GSM modem GPS handset

A. MICROCONTROLLER

Example ATMEG 328DIP entirely fits of laughter a lot of uses, since automotive industry and house mechanization appliance, industry. It is suits nicely for smart cards and also for the battery driven devices with small spending EEPROM recollection make it easy to approve microcontrollers to procedure in which stable

storage space of several parameter are essential. a smaller amount rate, disturb open management as well as give and short control utilization LPC231 micro- controllers base on a 32/16 bit ATMEG 328DIP hardware with valid-instance emulation still as constituted sketch maintain that assemble the microcontroller with 16kB, 32kB, 64kB, 128kB, 256kB and 512 kB of embedded elevated rate spark recollection [11].

The GPS unit's hardware interfaces are designed in such way that they should meet specification of NMEA [National Marine Electronics Association] necessities. The GPS handset provides information in NMEA 0183design through a 1Hz revise charge. In general communication usual by GPS is in NMEA communication set-up and NMEA procedure which is mainly normally used is NMEA0183 procedure [9].

B. IR SENSOR

The IR- sensor consist of Transmitter at one side of the track and another side (opposite) Receiver will be adjusted in site, all four IR-sensors that track the signals are placed if any emergency vehicle cross over the signal.

The IR sensor disconnect the incite and sends the information to the controller, when it detects the emergency vehicle crossing over the IR- line, sends the message to concerned Traffic controller leading to glowing of the LED.

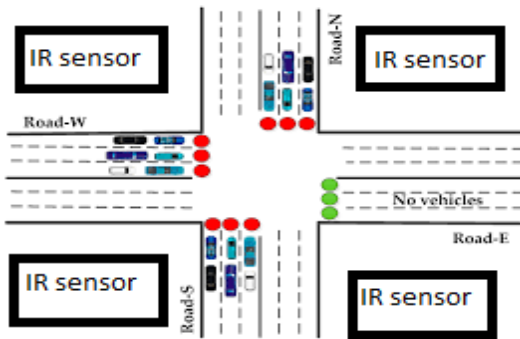
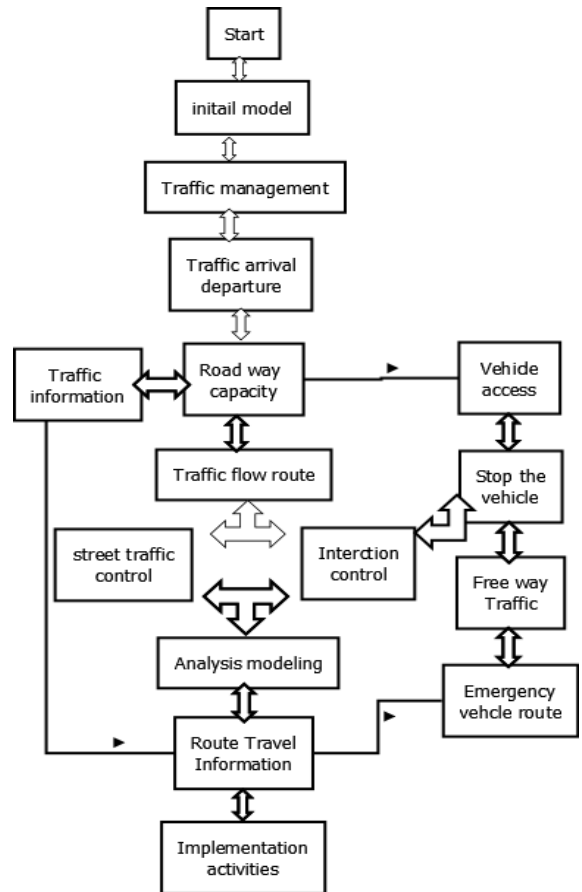


Fig 3. IR sensor Tracking System

C. FLOW CHART:-



IV. RESULT AND DISCUSSION

Comprehensive scheme as shown in fig.4 contain entire relationship of 32-bit ATMEG 328DIP CPU next to among GSM and GPS scheme. By this arrangement it becomes simple to maintain path of the emergent vehicle. If the objective reside in any position information can without problems live traced elsewhere.

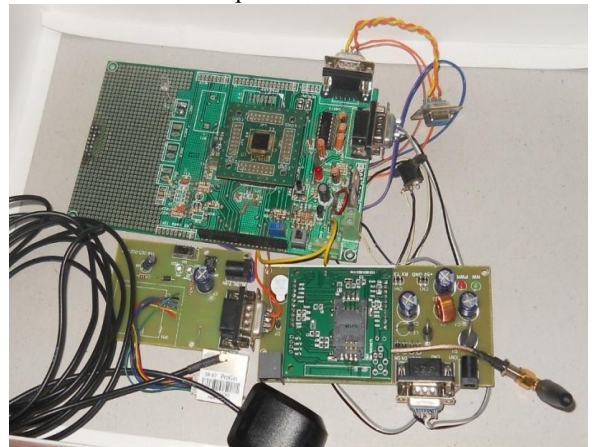




Fig. 4: tracking system.

TABLE I. PERFORMANCE COMPARISON

Existing system	Platform / Technology	Methodology	Comment
Bus Transportation System	PC based system	Location Tracking	Little Bit complex to design
Vehicle Tracking system	GPS and GPRS, Mobile based system	Easy to find vehicle Location	Easy to design
Ambulance monitoring system	and GPRS, Mobile based system	Easy to identify location	User friendly



Fig. 5: Message received by controller

CONCLUSIONS

The vehicles track method is suitable more and more significant in outsized city as fit as the interchange pointer which is other protected than extra system. It is from top to bottom included so that once it is implement in all interchange regulator, then it is likely to path an developing motor vehicle anytime beginning where. It has valid-instant potential, up to date information technology simultaneously and capable to type a valid-instant perfect, effective interchange gesture method. This method have various compensation such as huge

potential in discovery, large region collection, small process expenses, valuable in provided that suitable in order about a urgent situation motor vehicle and unproblematic to use in motor vehicle travel administration. Thanks to Anaswara R Lakshmi S

REFERENCES

- [1]. J. Saranya, J. Selvakumar, "Implementation of children tracking system on android mobile terminals," International Conference on Communication and Signal Processing, April 3-5, 2013.
- [2]. Swati Chandurkar, Sneha Mugade, Sanjana Sinha, Pooja Borkar, "Implementation of real time bus monitoring and passenger information system," International Journal of Scientific and Research Publications, Vol. 3, Issue 5, May 2013.
- [3]. Gangadhar, M. Madhu, M. S. Pushpalata, S. "Vehicle tracking and monitoring by ARM 7," SSRG International Journal of Electrical and Electronics Engineering (SSRG-IJEEE), Volume 1, Issue 4, June 2014.
- [4]. abid khan, ravi mishra, "gps-gsm primarily based monitoring gadget," global magazine of engineering developments and technology, vol. 3, trouble 2, pp: 161-164, 2012.
- [5]. Abid Khan & Ravi Mishra, —GPS – GSM Based Tracking Systeml, International Journal of Trends and Technology, ISSN: 2231 – 5381, Volume 3, Issue 2, 2012.
- [6]. Car Security and Tracking System with Position, Route, and Speed Calculation, EDP Topic VG06, 2011.
- [7]. M. A. Hannan, A. M. Mustapha, A. Hussain, H. Basri, "Intelligent bus monitoring and management system," Proceeding of the World Congress on Engineering and Computer Science, Vol. II, October 24-26, 2012.
- [8]. Jie Ding, Rui Wang and Xiao Chen. "Performance modeling and evaluation of real-time traffic status query for intelligent traffic systems." Communications (APCC), 2016 22nd Asia-Pacific Conference on. IEEE, 2016.
- [9]. subhadeep sarkar and sudip misra. "theoretical modelling of fog computing:a green computing paradigm to support iot packages." iet networks (volume: five, problem: 2, 3 2016). iet, 2016.
- [10]. Subhadeep Sarkar, Subarna Chatterjee and Sudip Misra. "Assessment of the Suitability of Fog Computing in the Context of Internet of Things." IEEE Transactions on Cloud Computing (Volume: PP, Issue: 99).IEEE, 2015.
- [11]. Qihui Wu, Guoru Ding, Shuo Feng, Zhiyong Du, Jinlong Wang and Keping Long. "Cognitive Internet of Things: A New Paradigm beyond Connection." IEEE Internet of Things Journal (Volume: 1, Issue: 2, April 2014). IEEE, 2014.